The Pisot Cross-Shell Residue:

A Reduction Lemma, Sharpness Witnesses, and an Exhaustive Quintic Execution of the ν-Criterion

with a three-run replication of the relational-charge signature ledger

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Session note; all computations exact, artifact manifest in Appendix A.

Abstract

This note consolidates one live verification session against the relational-charge framework of [1]. Contributions, each carrying its epistemic tag: (1) a *hypothesis-necessity witness* $Z^* = x^4 - 3x^2 + 1 = (x^2 - x - 1)(x^2 + x - 1)$ showing that both the irreducibility hypothesis of the modulus-pinning theorem and the disjoint-modulus clause of its mixed form are necessary `[FORCED]`; (2) a *real-pair ν-reduction lemma*: every real×non-real coherence question is decided by the shell scan of Rat_p alone `[FORCED]`, which forces relational inertness for every Pisot number with at most one pair of non-real conjugates `[FORCED]` and locates the first genuinely open cross-shell residue of [1, Cor. 7.16] at the quintic two-pair pattern; (3) the *first execution of that residue*: an exhaustive exact census of the 3125 monic quintics with coefficients in $[-2, 2]$ certifies 83 Pisot quintics, of which 67 realize the two-pair pattern (all on distinct shells), and the composed-square instrument C_2 returns exactly $\{\Phi_1^{20}\}$ on all 67: *zero mirrored cross-shell classes* (`[FORCED]` per instance; `[COMPUTED]` over the box); (4) companion live replications: the degree-12 Salem census (378 twist-classes, tally 39/257/45/37, 37/37 inert), a Pisot-quartic sweep (103/103 scans $\{\Phi_1^4\}$, emission-gap probe unfalsified), and a third run of signature ledger M in PARI/GP with *engine-native selection* and two in-engine decision paths asserted equal on all 237 signatures (474 scan executions, 0 disagreements) `[COMPUTED]`; (5) two reproducible verification-harness rules for PARI/GP decks, discovered by a pinned-tally failure and codified `[FORCED]`. Next steps are enumerated with falsifiable predictions, including the general Pisot-inertness conjecture `[OPEN]` in its multiplicative-torsion formulation. *v1.1 addendum:* the box extensions N2–N3 are executed and independently verified (§9): 431 Pisot quintics in $[-3, 3]^5$ with all 313 two-pair instances inert (P2 resolved: 0 mirrored classes), and 589 Salem twist-classes in $\{-2, \ldots, 2\}^6$, all inert (P5, P6 confirmed); zero falsifiers, nothing promoted.

1 Scope, engines, and discipline

All statements below concern the objects of [1]: for a monic $p \in \mathbb{Z}[x]$ with $p(0) \neq 0$ and roots $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$, the *angle* $t(\alpha) \in \mathbb{R}/\mathbb{Z}$ is the normalized argument, two roots are *coherent*, $\alpha \approx \beta$, iff $t_{\mathrm{rel}}(\alpha, \beta) = t(\alpha) - t(\beta) \in \mathbb{Q}/\mathbb{Z}$, a *shell* is a level set of the modulus, and the *relational type* is the partition of the root set into coherence classes together with the class offsets. The *ratio object* is

$$\mathrm{Rat}_p(x) \;=\; \mathrm{prim}\ \mathrm{Res}_y\big(p(y),\, p(xy)\big) \in \mathbb{Z}[x].$$

Two engines were used, matching the ledger-M standard of [1]: SYMPY 1.14.0 on Python 3.12.3 (arbitrary-precision Python integers), and PARI/GP 2.15.4 on GMP 6.3.0 (amd64, gcc 13.2.0 build).

No floating-point value crosses any decision boundary: every equality, divisibility, and inequality below is settled by exact division in $\mathbb{Z}[x]$, Sturm counts at rational endpoints, or rational-interval certificates. Where residues are pruned modulo the primes $10^9{+}7$ and 998244353, pruning discards *negatives only*; every positive is confirmed by exact division over $\mathbb{Z}$. Tags follow house convention: `[FORCED]` (machine-verified theorem / exact per-instance certificate), `[COMPUTED]` (exact computation over a stated finite range), `[PLAUSIBLE]`, `[DECLARED]`, `[OPEN]`.

2 The scan as a decision procedure

Proposition 2.1 (ratio object). *Let p be monic of degree n with $p(0) \neq 0$ and roots $\alpha_1, \dots, \alpha_n$ (with multiplicity). Then*

$$\operatorname{Res}_y\big(p(y), p(xy)\big) = \big((-1)^n p(0)\big)^n \prod_{i,j}\Big(x - \frac{\alpha_j}{\alpha_i}\Big),$$

so Rat_p has degree exactly n^2, root multiset the n^2 ordered ratios, and integer coefficients. Moreover the multiplicity of $(x-1)$ in Rat_p equals $\sum_{\text{distinct roots}} m_\alpha^2$; in particular it equals n iff p is squarefree. `[FORCED]`

Proof. With $f = p(y)$ monic and $g = p(xy)$ of y-degree n, $\operatorname{Res}_y(f, g) = \operatorname{lc}(f)^{\deg g} \prod_{p(\alpha_i)=0} p(x\alpha_i)$ and $p(x\alpha_i) = \prod_j (x\alpha_i - \alpha_j) = \alpha_i^n \prod_j (x - \alpha_j/\alpha_i)$; collect $\prod_i \alpha_i^n = ((-1)^n p(0))^n \neq 0$. The diagonal count is $\#\{(i,j) : \alpha_i = \alpha_j\} = \sum m_\alpha^2$. □

Lemma 2.2 (contact criterion). *Suppose $|\alpha| = |\beta|$. Then $\alpha \approx \beta$ iff α/β is a root of unity; and if α/β is a primitive M-th root of unity then $\Phi_M \mid \operatorname{Rat}_p$ in $\mathbb{Z}[x]$ (Gauss). Conversely $\Phi_M \mid \operatorname{Rat}_p$ exhibits a primitive M-th root of unity among the ratios.* `[FORCED]`

Lemma 2.3 (completeness bound). *For every $M \geq 1$, $2\varphi(M)^2 \geq M$, with equality iff $M = 2$. Hence $\Phi_M \mid P$ forces $M \leq 2(\deg P)^2$; the complete contact scan of Rat_p is a finite decision procedure with bound $2n^4$, entirely in $\mathbb{Z}[x]$.* `[FORCED]`

Proof. Odd prime power: $\varphi(p^b) = p^{b-1}(p-1) \geq p^{b-1}\sqrt{p} = p^{b-\frac{1}{2}} \geq p^{b/2}$, using $p - 1 \geq \sqrt{p} \iff p^2 - 3p + 1 \geq 0$ for $p \geq 3$. Two-part, $a \geq 1$: $\varphi(2^a) = 2^{a-1} \geq 2^{(a-1)/2}$, equality iff $a = 1$. Multiply: $\varphi(M) \geq \sqrt{M/2}$, tight only at $M = 2$. The equality case was swept exactly to $M \leq 2 \cdot 10^5$: no violations, unique tight case $M = 2$. `[COMPUTED]` □

Remark 2.4 (ν-criterion, two-part caveat)**.** For a cross-shell pair set $\mu = \alpha/\beta$ and $\nu = \mu/\bar{\mu}$; then $t(\nu) = 2\,t_{\text{rel}}$, so *rationality* of the offset is decided exactly while its *order* is read only up to the 2-part ([1, Rem. 7.6]).

Engine corroboration of §2 on the canonical ledger (signatures A–X of [1]) reproduced every row, including the Lehmer scan ($\deg \operatorname{Rat} = 100$, bound 20000, $\{\Phi_1^{10}\}$, 0.89 s against the deposit's 1.1 s), the mixed scan $\operatorname{Rat}_{L,\beta_4}$ (deg 40, empty), the Salem certifications of β_4, S_6, S_8, L by trace-polynomial Sturm patterns $(1, 0, n/2-1)$, the two-route identity $\operatorname{charpoly}(C \otimes C^{-1}) = \operatorname{Rat}_{\beta_4}$, and the Kronecker-square factorization $\operatorname{charpoly}(C_{x^4-x+1} \otimes C_{x^4-x+1}) = S_6^2 \cdot (x^4 + 2x^2 - x + 1)$ with the quartic factor equal to the ψ^2-image. Total: 1.5 s. `[FORCED]` per instance. Artifact: `pinning_scan_scratch.py`.

3 Modulus pinning and its sharpness

Theorem 3.1 (modulus pinning; [1, Thm. 7.13], proof included for self-containment)**.** *Let $p \in \mathbb{Z}[x]$ be irreducible with a root α_0 whose modulus is attained by no other root. Then no two distinct roots* *of p differ by a root-of-unity factor. Mixed form: if $q \neq p$ is irreducible and no root of q has modulus $|\alpha_0|$, then no torsion ratio exists between a root of p and a root of q.* `[FORCED]`

Proof. Suppose $\alpha \neq \beta$, $p(\alpha) = p(\beta) = 0$, $\alpha = \zeta\beta$ with $\zeta^M = 1$. Transitivity of $G = \mathrm{Gal}$ of the splitting field (the *only* use of irreducibility) supplies σ with $\sigma\alpha = \alpha_0$. Since $(\sigma\zeta)^M = 1$, $\sigma\zeta$ is torsion, hence $|\sigma\zeta| = 1$; thus $|\sigma\beta| = |\alpha_0|$, forcing $\sigma\beta = \alpha_0 = \sigma\alpha$ and $\alpha = \beta$. For the mixed form, work in the splitting field L of pq; restriction $\mathrm{Gal}(L/\mathbb{Q}) \twoheadrightarrow \mathrm{Gal}(K_p/\mathbb{Q})$ lifts transitivity, and $\sigma\beta = \sigma\alpha/\sigma\zeta$ is a q-root at the excluded modulus. □

Remark 3.2 (proof audit)**.** (i) The complex modulus is *not* Galois-equivariant and the proof never needs it to be; only the Galois-stable predicate "is torsion" plus $|\cdot| = 1$ for torsion enters. (ii) The pinned root need not participate in the offending pair. (iii) "Uniquely attained", not "maximal", is the hypothesis; a Salem polynomial carries two pins (τ and $1/\tau$). Corollaries [1, 7.14, 7.15] (Salem inertness; no cross-locking of distinct Salems) follow as in the parent paper and were re-derived and engine-checked.

Proposition 3.3 (hypothesis-necessity witness Z^*)**.** *Let $f_1 = x^2 - x - 1$, $f_2 = x^2 + x - 1$, and $Z^* = f_1 f_2 = x^4 - 3x^2 + 1$. Each f_i is irreducible and* both *of its root moduli (φ and φ^{-1}) are uniquely attained within that factor, so per-factor pinning holds; yet Z^* carries the torsion ratio -1 between the roots φ (of f_1) and $-\varphi$ (of f_2) at the shared modulus φ. Consequently: (i) irreducibility cannot be dropped from Theorem 3.1; (ii) the disjoint-modulus clause cannot be dropped from its mixed form. The complete scan confirms $\mathrm{Rat}_{Z^*} \rightsquigarrow \{\Phi_1^4, \Phi_2^4\}$.* `[FORCED]`

Proof. f_1 has roots $\varphi, -1/\varphi$; f_2 has roots $1/\varphi, -\varphi$. Moduli per factor: $\{\varphi, \varphi^{-1}\}$, each once. Across factors, $\varphi/(-\varphi) = -1$ is torsion at the shared modulus φ. The same-polynomial form fails only through transitivity (absent for the reducible product); the mixed form fails only through its shared-modulus clause. The signature $\{\Phi_1^4, \Phi_2^4\}$ (Φ_1 from the diagonal, Φ_2 from the two antipodal pairs, ordered) is engine-confirmed in both stacks. □

Remark 3.4. Z^* complements the parent paper's sharpness example $x^4 - 2$ (which shows the pinning *hypothesis* necessary for an irreducible input): together the two witnesses isolate every hypothesis of Theorem 3.1 and its mixed form. Candidate placement: a remark beside [1, Ex. 7.19].

4 The real-pair ν-reduction; inertness below two non-real shells

Lemma 4.1 (real-pair ν-reduction)**.** *Let O be a conjugation-closed set of nonzero algebraic numbers, $\beta \in O$ real, $\alpha \in O$ non-real. Then*

$$\alpha \approx \beta \iff \alpha/\bar{\alpha} \text{ is a root of unity.}$$

If O is the root set of p, then $\alpha/\bar{\alpha}$ is a root of Rat_p*; hence every real$\times$non-real coherence question — same shell or cross shell — is decided by the complete contact scan of* Rat_p *alone.* `[FORCED]`

Proof. $t(\beta) \in \{0, \frac{1}{2}\}$, so $t(\alpha) - t(\beta) \in \mathbb{Q} \iff t(\alpha) \in \mathbb{Q} \iff 2t(\alpha) \in \mathbb{Q}$. The number $\alpha/\bar{\alpha}$ is unimodular with $t(\alpha/\bar{\alpha}) = 2t(\alpha)$; unimodular with rational angle is exactly torsion (Lemma 2.2). Since $\bar{\alpha} \in O$, the quantity is an ordered ratio, i.e. a root of Rat_p. [1, Ex. 7.17(iii)] is the β_4 instance of this statement. □

Theorem 4.2 (Pisot inertness below two non-real shells)**.** *Every Pisot number whose minimal polynomial has at most one pair of non-real conjugates — in particular every Pisot number of degree ≤ 4, and every quintic Pisot of pattern $(\theta, r_1, r_2, \lambda, \bar{\lambda})$ — is relationally inert: its type is one rational block containing exactly the real conjugates, plus singletons $\{\lambda\}, \{\bar{\lambda}\}$ when the pair exists.* `[FORCED]`

Proof. The dominant root $\theta > 1$ pins its modulus (all other conjugates lie in the open unit disk), so Theorem 3.1 excludes every torsion ratio between distinct conjugates; in particular $\lambda/\bar{\lambda}$ is not torsion, so by Lemma 4.1 no real×non-real pair is coherent, and $\lambda \not\approx \bar{\lambda}$ within the pair's shell. Real×real pairs have angle difference in $\frac{1}{2}\mathbb{Z} \subset \mathbb{Q}$ and form the rational block. With at most one non-real pair there is no non-real cross-shell pair left to decide. □

Corollary 4.3. *The open cross-shell residue of [1, Cor. 7.16] is empty through the patterns covered by Theorem 4.2. Its first live case is the quintic two-pair pattern $(\theta, \lambda_1, \bar{\lambda}_1, \lambda_2, \bar{\lambda}_2)$.* `[FORCED]`

5 The residue at two non-real shells

Throughout this section p is an irreducible Pisot quintic of two-pair pattern; write $u_i = \lambda_i/\bar{\lambda}_i$ (unimodular, and by Proposition 2.1 a root of Rat_p), and $\mathrm{Rat}_p^\circ = \mathrm{Rat}_p/(x-1)^5$.

Proposition 5.1 (ν inventory; level-2 formulation)**.** *For distinct shells $|\lambda_1| \neq |\lambda_2|$:*

$$\lambda_1 \approx \lambda_2 \iff u_1\bar{u}_2 \text{ torsion}, \qquad \lambda_1 \approx \bar{\lambda}_2 \iff u_1u_2 \text{ torsion}.$$

Hence a mirrored cross-shell class exists iff two unimodular roots of Rat_p° differ multiplicatively by a root of unity. The degenerate case $u_1 = u_2$ (offset in $\frac{1}{2}\mathbb{Z}$) forces a repeated root of Rat_p° and is detected by its squarefreeness. `[FORCED]`

Proof. $t(u_1\bar{u}_2) = 2(t_1 - t_2)$ and $t(u_1u_2) = 2(t_1 + t_2)$; rationality transfers both ways (Remark 2.4 affects only the order, not the decision). Both u_2 and $u_2^{-1} = \bar{u}_2$ are roots of Rat_p. If $u_1 = u_2$ as values, two distinct ordered ratios coincide. □

Proposition 5.2 (no level-2 pin)**.** *In the two-pair pattern with distinct shells, the modulus multiset of Rat_p° is*

$$\Big\{\tfrac{\theta}{|\lambda_1|}, \tfrac{\theta}{|\lambda_2|}, \tfrac{|\lambda_1|}{\theta}, \tfrac{|\lambda_2|}{\theta}\ (\times 2 \text{ each}),\ \tfrac{|\lambda_1|}{|\lambda_2|}, \tfrac{|\lambda_2|}{|\lambda_1|}\ (\times 4 \text{ each}),\ 1\ (\times 4)\Big\},$$

every value attained at least twice (coincidences only merge classes). Hence Theorem 3.1 is inapplicable one level up: no transport argument closes the residue, and a scan is genuinely required. `[FORCED]`

Proposition 5.3 (the S^*/C_2 instrument)**.** *Let S^* be the product of the self-reciprocal irreducible $\mathbb{Z}$-factors of Rat_p° that carry at least one unimodular root (certified per factor by the Sturm count of the exact trace square on $(-2, 2)$), $d = \deg S^*$, and*

$$C_2(x) \;=\; \mathrm{Res}_y\big(S^*(y),\, y^d S^*(x/y)\big).$$

Then: (a) the minimal polynomial of any unimodular algebraic number $u \neq \pm 1$ is self-reciprocal (its Galois orbit is inversion-closed, since $u^{-1} = \bar{u}$ lies in it), so $u_1, \bar{u}_1, u_2, \bar{u}_2$ are roots of S^*; (b) C_2 is monic of degree d^2 with root multiset all ordered products of S^*-roots; (c) negative certificate*: if the complete contact scan of C_2 returns exactly $\{\Phi_1^d\}$, then no product of two S^*-roots is a root of unity; in particular $u_1\bar{u}_2$ and u_1u_2 are not torsion and no mirrored cross-shell class exists; (d) the forced Φ_1-multiplicity is exactly d (each root pairs uniquely with its inverse; ± 1 are excluded from S^* by level-1 pinning), so any excess or any Φ_M, $M \geq 2$, is a signal.* `[FORCED]`

Proposition 5.4 (shell detector)**.** *For the two-pair pattern, the number of unimodular roots of Rat_p° equals 4 iff $|\lambda_1| \neq |\lambda_2|$ and 12 iff $|\lambda_1| = |\lambda_2|$; the same-shell case is already closed by level-1 pinning (all its cross pairs are ordered ratios of conjugates).* `[FORCED]`

6 Exhaustive quintic execution

Certification (stage 1). Over the box of monic quintics $x^5 + ax^4 + bx^3 + cx^2 + dx + e$, $(a, \ldots, e) \in [-2, 2]^5$ (3125 candidates), each candidate passes, in order: $e \neq 0$; rejection of ±reciprocal coefficient vectors (an irreducible polynomial with a unimodular root is ±reciprocal, so this both removes the only carriers of circle roots and guarantees the disk certificates terminate — the Salem-boundary lesson of the quartic sweep, where $|\lambda| = 1$ makes $\theta r = d$ exactly and interval refinement cannot separate); irreducibility; the Sturm real pattern (exactly one root in $(1, \infty)$, none in $(-\infty, -1]$); and rational-rectangle disk certificates for each non-real root via indexed `CRootOf` rational refinement (indexed construction is required because solvable quintics return radical forms). Tallies:

reject: $e = 0$	625	reject: ±reciprocal	50
reject: reducible	638	reject: real pattern	1318
reject: disk certificate	411	**certified Pisot**	**83**
patterns: real5 = 0; mixed = 16; two-pair = **67**			

Decision (stage 2). All 83 ratio scans ($\deg \mathrm{Rat} = 25$, bound 1250, 53 totient-filtered candidates) returned exactly $\{\Phi_1^5\}$, as Theorem 3.1 forces; the 16 mixed instances are thereby fully typed (Theorem 4.2). On the 67 two-pair instances: the shell detector read 4 on *every* instance (all distinct-shell; no same-shell case occurred in the box); Rat_p° was squarefree and in fact *irreducible of degree* 20 *on all* 67 (so $S^* = \mathrm{Rat}_p^\circ$, $\deg C_2 = 400$ uniformly); the 67 complete C_2 scans (PARI/GP trial division over the master list of 790 cyclotomic candidates with $\varphi(m) \leq 400$, largest $m = 1680$) returned exactly $\{\Phi_1^{20}\}$ each, with two instances independently re-derived end-to-end in SYMPY (composed square by resultant, scan by trial division with two-prime negative pruning), in exact agreement.

Theorem 6.1 (quintic box result)**.** *Every Pisot quintic with coefficients in* $[-2, 2]$ *is relationally inert: its type is the rational block of its real conjugates plus non-real singletons. In particular, across the first* 67 *live instances of the cross-shell residue,* no mirrored cross-shell class exists*, and no degenerate* $\frac{1}{2}\mathbb{Z}$*-offset coherence occurs (*Rat_p° *squarefree,* 67/67*).* `[FORCED]` *per instance; box coverage* `[COMPUTED]`*.*

Observation 6.2. Rat_p° irreducible on 67/67 instances is the Galois-generic expectation (transitive action on ordered root pairs); factored instances are predicted for special Galois groups and would shrink S^* and C_2. `[PLAUSIBLE]`

Wall times: certification 64 s; stage 2 total 69 s (PARI/GP C_2 block 39.9 s; SYMPY two-route samples 10.3 s and 9.1 s). The smallest two-pair instance encountered first in enumeration order is $x^5 - 2x^4 - 2x^3 - 2x^2 - 2x - 2$.

7 Companion live replications

7.1 Degree-12 Salem census ([1, Thm. 7.10])

Independent re-implementation (deposit scripts not consulted): Burnside 729 vectors, 27 twist-fixed, 378 orbits; cascade tally exactly 39 (±1 root) / 257 (trace-Sturm) / 45 (reducible) / **37** Salem; all 37 complete scans ($\deg \mathrm{Rat} = 144$, bound 41472, totient filter collapsing to 290 candidates, largest $m = 630$) returned $\{\Phi_1^{12}\}$; twist-invariance of Rat verified on a live representative. 60 s against the deposit's 93 s. `[FORCED]` per instance.

7.2 Pisot-quartic sweep and the emission-gap probe

Box $[-3,3]^4$ (2401 candidates): 103 certified Pisot quartics (102 complex-pair, 1 totally real); 103/103 complete scans returned exactly $\{\Phi_1^4\}$, the Theorem 3.1 prediction. `[COMPUTED]` over the box. The totally-real subfamily is charge-admissible: all angles lie in $\{0, \frac{1}{2}\}$ and $M(p) = \theta$ (only θ lies outside the closed unit disk), so the corpus emission-gap theorem $M \in \{1\} \cup [\varphi, \infty)$ predicts $\theta \geq \varphi$. The unique totally-real hit, $x^4 - 3x^3 - 2x^2 + 2x + 1$ ($\theta \approx 3.390$, display value only), satisfies $\theta > \varphi$ by exact certificate: the probe does not falsify the gap. The smallest complex-pair instance is $x^4 - x^3 - 1$ ($\theta \approx 1.3803$).

7.3 Signature ledger M, third run

Independence audit:

run	engine / kernel	decision path	instance selection	operator
1	SYMPY 1.14.0 / Python ints	trial division	Python cascade + sweeps	same
2	PARI/GP 2.15.4 / GMP 6.3.0	`factor+poliscyclo`	fed from run 1	same
3	PARI/GP 2.15.4 / GMP 6.3.0	both, asserted equal	PARI/GP-native census; re-typed canonicals	same

Run 3 re-derives the census *selection* inside PARI/GP (Burnside quotient, ± 1 check, trace-square Sturm, irreducibility): BURNSIDE $(729, 27, 378)$, CASCADE $(39, 257, 45, 37)$, selection set identical to run 1 (37/37). Deck outcome: 237/237 signatures agree (canonical 14 including Z^*, the mixed pair, and [1, Ex. 7.18]; census 37; quartic Pisots 103; quintic Pisots 83), each decided by two in-engine paths for 474 scan executions with 0 path disagreements; ledger-O certifications 4/4. Wall time 1.8 s. Independence achieved: engine, bignum kernel, decision algorithm, selection pipeline. Not achieved and disclosed per [1, §9]: the session operator and the shared family definitions. Consistency claim `[COMPUTED]`; each per-instance signature independently `[FORCED]`.

8 Verification-harness findings

Both discovered by a pinned-tally failure (the truncated run produced the internally consistent but wrong cascade $(108, 270, 0, 0)$) and reproduced in a minimal probe. `[FORCED]`

H1. PARI/GP terminates top-level statements at newline: a function body split across physical lines without braces silently truncates. Rule: multi-line PARI/GP definitions must be brace-wrapped; single-line otherwise.

H2. `gp` exits 0 after load errors, emitting `***` diagnostics to stderr only ("skipping file"). Rule: harnesses must treat any non-benign `***` stderr line as fatal.

H3. Positive pattern: the dual-path $\mathrm{sig}(P) = \mathrm{scan}_{\mathrm{td}}(P) \stackrel{!}{=} \mathrm{scan}_{\mathrm{fp}}(P)$ assertion (trial division vs. factorization) as the in-engine standard for PARI/GP-side scans.

H4. `[DECLARED]` Container-class observation, disclosed rather than promoted: one execution container reaped backgrounded processes at call teardown nondeterministically (the N3 driver survived only because its launch call outlived it; the N2 driver was killed mid-import with an empty log); a second container of the same family did *not* reproduce it (five harness-tracked background runs, up to 15 minutes, all completed). Binding pattern regardless: foreground execution with checkpointed, restart-safe drivers.

9 Box extensions, executed (v1.1 addendum)

Both box extensions of §10 (N2, N3) were executed post-archive and independently re-verified end-to-end (zero-shot verification session, 2026-07-02: manifest integrity, full regeneration of every pinned artifact — byte-identical, also under Python 3.11.15 — and an adversarial referee pass over every `[FORCED]`-tagged certificate). The corrections below are recorded per the project discipline that load-bearing corrections are pinned rather than papered over.

9.1 N2: quintic box $[-3,3]^5$

Census. 16807 candidates; 14406 with $e \neq 0$; exact certificate partition 862 (interior count 4 on every non-degenerate path) / 10942 (exact count $\neq 4$) / 2602 (all paths degenerate). The 862 pass the Sturm and irreducibility gates down to **431** certified Pisot quintics (313 two-pair, 118 mixed, 0 totally real); the $[-2,2]^5$ restriction reproduces §6 exactly (83/67/16, row-for-row). The 2602 degenerate-chain candidates were adjudicated exactly by two *independent* methods (certified rational root enclosures; scaled Schur–Cohn sandwich at rational radii straddling 1), agreeing instance-for-instance: 2594 reducible, 8 irreducible with interior count in $\{2,3\}$, 0 Pisot — the census is complete. `[COMPUTED]` Disclosure: the four-path certificate chain $\{\mathrm{SC}, \mathrm{RH}\}$ on $\{p(x), -p(-x)\}$ collapses to *two* independent methods (both transforms commute with $x \mapsto -x$; verified with 0 exceptions and 0 cross-path disagreements over all 14406 candidates), so per-instance certificates carry at most two independent computations. `[FORCED]`

Decision. 313/313 complete C_2 scans returned exactly $\{\Phi_1^{20}\}$, with Φ_2-multiplicity 0 and no higher cyclotomic content anywhere; the 67 instances shared with §6 agree row-for-row with run 1. `[FORCED]` per instance; `[COMPUTED]` over the box. No same-shell two-pair instance occurred (P8 extended to $[-3,3]^5$).

Remark 9.1 (shell-detector semantics, corrected)**.** Proposition 5.4 counts unimodular roots *with multiplicity* (4 vs. 12). An implementation reading the trace-square Sturm count on $(-2,2)$ counts *distinct* values instead; in the merged case the 12 ratios provably collapse to at most 4 distinct trace values (the cross ratios pair off), so a merged branch keyed to the distinct count 6 is unreachable, and a torsion-merged configuration can present the distinct-shell count silently. The merged case is nonetheless excluded on every instance by the joint certificate actually asserted — distinct-shell count, Φ_2-free, no higher cyclotomic content, no count anomaly — since any silent collapse forces torsion ratios that the C_2/torsion scan exhibits. In the $[-2,2]^5$ run the verified squarefreeness of Rat_p° already equates the two counts. `[FORCED]`

9.2 N3: census family $\{-2,\ldots,2\}^6$

15625 vectors, 125 twist-fixed, **7875** classes (P5, asserted in-run). Cascade, with `[DECLARED]` category labels (a three-way partition, not claimed comparable to the archived cascade tallies): 6548 straddle-fail / 738 straddle-pass-reducible / **589** Salem classes. The $\{-1,0,1\}^6$ restriction reproduces the archived census exactly (378 classes, 37 Salem, 37/37 inert). All 589 scans returned exactly $\{\Phi_1^{12}\}$ with the Φ_2-impossibility assertion holding per instance: the corroboration base of the modulus-pinning theorem on Salem classes grows $37 \to 589$ with zero falsifiers (P6). `[FORCED]` per instance; `[COMPUTED]` over the box. Correction, pinned: the twist-exclusivity shortcut ("the two class members cannot both straddle") fails at Sturm endpoints — 160 classes both straddle, each provably with $P(\pm 1) = 0$, hence reducible — so exclusivity holds in the weaker, sufficient form "both straddle $\Rightarrow$ both reducible." `[FORCED]`

10 Next steps and predictions

10.1 Operational fronts

N1. Symmetrize C_2 coverage: full SYMPY-side scans of all 67 composed squares ($\approx 67 \times 10\,\mathrm{s} \approx 11\,\mathrm{min}$). *Executed:* 67/67 returned $\{\Phi_1^{20}\}$; instance 1 additionally re-derived by a fully symbolic degree-400 resultant, identical to the Newton route.

N2. Quintic box extension to $[-3,3]^5$ (16807 candidates; certification $\approx 6\times$ stage-1 cost). *Executed:* §9.

N3. Census family extension $c \in \{-2,\ldots,2\}^6$: 15625 vectors; twist-fixed $5^3 = 125$; Burnside gives *exactly* $(15625+125)/2 = 7875$ twist-classes (forced arithmetic); cascade tallies to be computed. *Executed:* §9.

N4. Degrees 6–7: degree 6 contributes patterns $(\theta, r, \lambda_1, \bar\lambda_1, \lambda_2, \bar\lambda_2)$ (two-pair with a real spectator — machinery unchanged) and real-heavy patterns closed by Theorem 4.2; the first three-pair pattern appears at degree 7, and C_2 still suffices there since every pairwise ν is a product of two u's.

N5. Codify H1–H3 into the repository's cross-engine verification standard beside the engine version pins. *Partially executed:* H1–H2 and the ledger-M live replication folded into the parent deposit (v1.1, K-series); H4 recorded in §8.

N6. Fold Proposition 3.3 (beside [1, Ex. 7.19]) and Lemma 4.1 (beside [1, Rem. 7.6]) into a v1.1 of the parent paper; sharpen [1, Cor. 7.16] by Corollary 4.3.

N7. Provenance pass (Crossref/arXiv only, per house discipline) on: minimality of totally-real Pisot numbers, and multiplicative (in)dependence of conjugates / torsion quotients (the Dubickas-program direction referenced in the provenance block of [1]).

10.2 Predictions

Conjecture 10.1 (Pisot inertness, general)**.** *Every Pisot number is relationally inert: its type is the rational block of its real conjugates plus non-real singletons. Equivalently, for conjugate-ratio values $u_i = \lambda_i/\bar\lambda_i$ on distinct shells, $u_i u_j^{\pm 1}$ is never a root of unity.* `[OPEN]`

Status: `[FORCED]` for at most one non-real pair (Theorem 4.2); `[COMPUTED]`-empty on the first 67 live instances (Theorem 6.1) and on all 313 two-pair instances of $[-3,3]^5$ (§9); Proposition 5.2 shows transport cannot prove it, so a genuinely new mechanism (heights, multiplicative-relation bounds) is required.

ID	prediction	status	falsifier
P1	Conjecture 10.1 holds.	`[OPEN]`	any certified Pisot with a C_2-contact Φ_M, $M \geq 2$, or a Φ_1 excess / non-squarefree Rat_p°
P2	N2 yields 0 mirrored classes among all two-pair instances in $[-3,3]^5$.	`[COMPUTED]` (resolved: 0/313, §9)	a single hit in any further extension
P3	Every certified Pisot's Rat scan returns exactly $\{\Phi_1^{\deg p}\}$ in any box extension.	`[FORCED]` (Thm. 3.1)	— (a deviation would falsify the pinning theorem)
P4	Every totally-real Pisot number satisfies $\theta \geq \varphi$, equality only at $\theta = \varphi$ (degree 2).	emission-gap–implied; standalone `[OPEN]`	a totally-real Pisot with $\theta < \varphi$ (would falsify the gap on admissible objects)
P5	N3 Burnside count is exactly 7875 twist-classes.	`[FORCED]` (arithmetic; observed 7875)	—
P6	Every Salem class found by N3 is inert with scan $\{\Phi_1^{12}\}$.	`[FORCED]` ([1, Cor. 7.14]; observed 589/589)	—
P7	Two-pair instances with Rat_p° reducible (smaller S^*) exist for non-generic Galois groups.	`[PLAUSIBLE]`	exhaustive extensions showing irreducibility persists
P8	Same-shell two-pair Pisot quintics (shell detector = 12): none occurred in $[-2,2]^5$ or $[-3,3]^5$ (§9, Remark 9.1); existence elsewhere undetermined.	`[OPEN]`	(either outcome informative)

11 Provenance and citation discipline

Bibliographic pointers here are restricted to the corpus's own archived records (DOIs below) and to prior art *as attributed inside* [1, Rem. 7.20 and refs. 14–16]. No literature search was performed in-session; external attributions (torsion quotients of conjugates; totally-real Pisot minimality) are therefore inherited, not re-verified, and N7 schedules the Crossref/arXiv pass. All numerical displays marked "display value" carry no decision weight.

A Artifact manifest, engines, runtimes

Engines: SYMPY 1.14.0 / Python 3.12.3; PARI/GP 2.15.4 / GMP 6.3.0. Runtimes: canonical corroboration 1.5 s; census 60 s; quartic sweep 4 s; cross-engine run 2 0.5 s; quintic certification 64 s; quintic decision 69 s; ledger-M run 3 1.8 s. v1.1 additions (authoring / verification rerun): N2 scans 154/207 s; N3 census-and-scans 33/46 s; every v1.1 artifact regenerates byte-identically under both Python 3.12.3 and 3.11.15.

artifact	SHA-256
pinning_scan_scratch.py	9b9d95156532057b841d5f07b33428d1ab02a18dfe416fc3086427a1b444f090
front1_census.py	da3ef53064d33a3ce87ed37b652a4603dedee0451ff84e106d2098e34ef3d18f
front1_scans.py	d4ae1e47ebba0476291ce67ce63127c6050a60f21ad93019e0a5b2acdde5c7af
front2_pisot_sweep.py	bf5c81f26ecc4f54d16ee2e27901f1b86bb2389e6ec13c8e26acff0b2a73e145
front3_cross_engine.py	092b408806e9c8cbdec8bc4adb54737268bb12986cf7be1dc1202138b3e4372c
cross_live.gp	b07b8b3ac1f1466b8542d68fe32b7109964b0c8cfa691a3edc9d1c57d80b4ec5
nu_quintic_cert.py	283b588ea9b3ba70b97ac7e597d6bf3ecebfe5477e17065651f8b5d976bac083
nu_quintic_scan.py	9b5fccddcf40c1a70ab34cc8e3d662da8f4ad43b8720e1dad0783d208f3f7cf2
quintic_c2.gp	e199df7f6cb9d3e9f00b6bb55f520b254a6cdbeaa8d965f196dbab5223b879bb
quintic_pisots.json	49f3e1569d755416d86fb0ff95c337b86a1e5e3f5bee44e0bbd6a4bf5b1a969c
ledgerM_header.gp	ed4ca186fdf336004b185d7d9b745a6d31d5a12d5c9a1f6ab2256edf418aa4b2
ledgerM_live.gp	54fcd1488cdd37ab151877a98e0a3ec13f1f52dcf7799a44bbb47fc72e94349a
build_ledgerM.py	4b4ee6818f5b3dfed3706a88085f2dda72de36cc42fa85740b145f64abc5df8b
pisot103.json	fb349dc136e149718726df2e729184681b33fd22abb68585e077df7b0dfe792d

v1.1 artifact	SHA-256
filesfixed.zip (box-extension bundle)	8432455c20b07e9b6c2be5987484e8b720ceb27cefdaf7add6b34a28362ed3a1

The bundle's top-level manifest pins its four component manifests (`n1-symmetrize`, `n2-n3-box-extensions`, `n2-chain-audit`, `deposit-crossengine-fold`), the runbook, and the fix-log README. Changed pin, disclosed there: `pisot83.jsonl` re-pinned to the byte-exact output of its pinned generator (`aeff3c3d...`); mathematical fields unchanged, the dropped `certs` column preserved verbatim in `pisot_n2.jsonl`'s $[-2,2]^5$ restriction.

References

[1] J. Turner, *Relational Charge on the Spectral Semiring: Gauge Rigidity, Coherence Types, and a Reference-Free Parity Floor*, v1.0.0, Zenodo, doi:10.5281/zenodo.21121863; verification bundle v1.0.1, doi:10.5281/zenodo.21122410.

[2] Echo-S Studios, *math-research-pipelines* (repository and live site), `https://echo-s-studios.github.io/math-research-pipelines/`.